

Employee Attrition Prediction

A Big Data Analytics Case Study

Team Members

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Mitigating the Real-World Cost of Employee Attrition

Economic Impact

50–200%

Cost of replacing an employee as a percentage of annual salary, including recruitment, training, and lost productivity.

Operational Impact

Critical

High turnover disrupts team cohesion, reduces institutional knowledge, and negatively impacts service quality and innovation.

Proactive Solution

Predictive

Identify at-risk employees before they leave, enabling targeted retention strategies like career development and work-life balance improvements.

Project Goal

Build a robust machine learning model that accurately predicts employee attrition and identifies the key factors driving it, enabling data-driven HR interventions.

The Employee Dataset: A Good Fit for Attrition Analysis

Dataset Overview

Initial Size 74,610 records with 24 features

Features Comprehensive mix of demographic, organizational, and psychological factors

Why Good Fit Breadth of features allows holistic analysis beyond salary to explore behavioral and environmental drivers of attrition

Feature Types Demographic (Age, Gender), Organizational (Job Role, Level, Years), Psychological (Satisfaction, Balance, Performance)

Data Quality Issues & Handling

Issue 1: Missing Values

- Distance from Home: 2.56% missing (1,912 records)
- Company Tenure: 3.23% missing (2,413 records)

✓Imputed with median values to maintain distribution

Issue 2: Duplicate Records

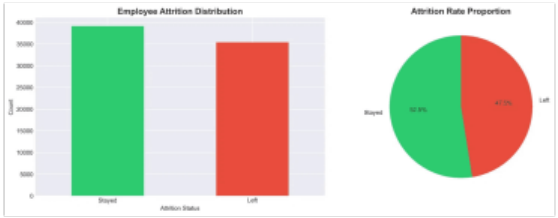
- 112 duplicate rows identified in dataset

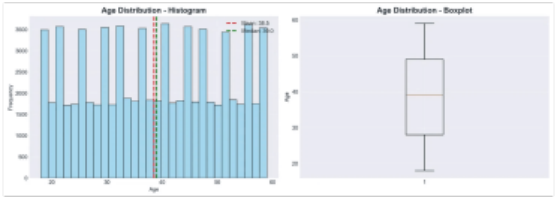
✓Removed duplicates to ensure data integrity

Final Dataset

74,498 unique, complete records

Initial Data Insights: A Balanced Attrition Problem







Attrition Distribution

52.5% vs 47.5%

Nearly balanced distribution between employees who stayed and those who left.

✓Balance reduces model bias toward majority class, enabling reliable predictions for both outcomes.

Age Distribution

Mean: 38.5 | Median: 39

Relatively uniform distribution across age groups (18–59 years).

✓Attrition is not limited to entry-level or retirement-age employees; organizational factors affect the entire workforce.

Monthly Income by Status

Similar Distributions

Income distributions for 'Left' and 'Stayed' employees are very similar.

✓Income is a factor but NOT the sole driver of attrition, reinforcing the need for multi-variate analysis.

Case Study: Identifying the High-Risk Employee Profile

Business Context

A large, multi-national corporation is struggling with high turnover and needs to understand **who is leaving** and **why** to implement effective retention programs. By analyzing employee data, we aim to predict attrition and identify actionable factors for targeted HR interventions.

1

Prediction: Can We Forecast Attrition?

Can we accurately predict which employees are most likely to leave in the next 12 months using classification models? This enables proactive identification of at-risk employees.

2

Drivers: What Factors Influence Attrition?

What are the most significant factors (e.g., Job Level, Work-Life Balance, Marital Status) that influence an employee's decision to leave? Understanding feature importance guides retention strategy priorities.

3

Segmentation: Can We Identify Employee Groups?

Can we identify distinct groups (clusters) of employees based on their characteristics to tailor retention strategies? Clustering enables customized HR interventions for different employee segments.

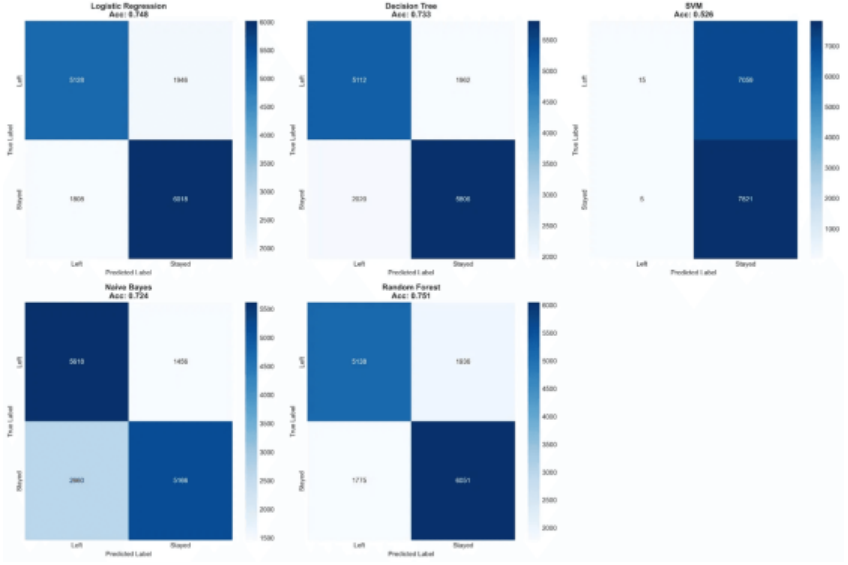
Supervised Learning: Predicting Attrition

Approach

We employed **Supervised Classification** to predict the binary outcome (Stayed/Left). This approach uses historical employee data with known attrition outcomes to train models that can predict future attrition.

Models Compared

- **Logistic Regression:** Baseline linear model for interpretability
- **Decision Tree:** Non-linear model, good for feature importance
- **Random Forest:** Robust ensemble method, typically high performance
- **Naive Bayes:** Probabilistic model based on conditional independence
- **Support Vector Machine (SVM):** Kernel-based model for complex boundaries



Model Selection Metrics

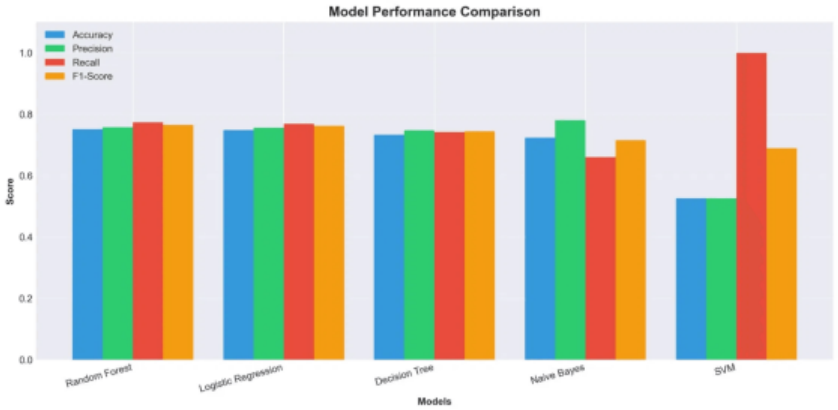
F1-Score: Balances Precision (correctly identifying those who will leave) and Recall (catching all those who will leave). Critical for minimizing false negatives.

AUC: Measures the model's ability to distinguish between classes across all possible classification thresholds.

Model Performance: Random Forest Delivers the Best Balance

Performance Metrics Comparison

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.751	0.758	0.773	0.765
Logistic Regression	0.748	0.756	0.769	0.762
Decision Tree	0.733	0.747	0.742	0.745
Naive Bayes	0.724	0.780	0.660	0.715
SVM	0.526	0.526	0.999	0.689



Statistical Justification

Random Forest achieved the highest F1-Score (0.765) and strong Recall (0.773), making it most effective for identifying at-risk employees.

SVM shows high Recall (0.999) but low Precision (0.526), causing too many false positives—impractical for targeted intervention.

Conclusion: Random Forest provides the optimal balance of predictive metrics for business application.

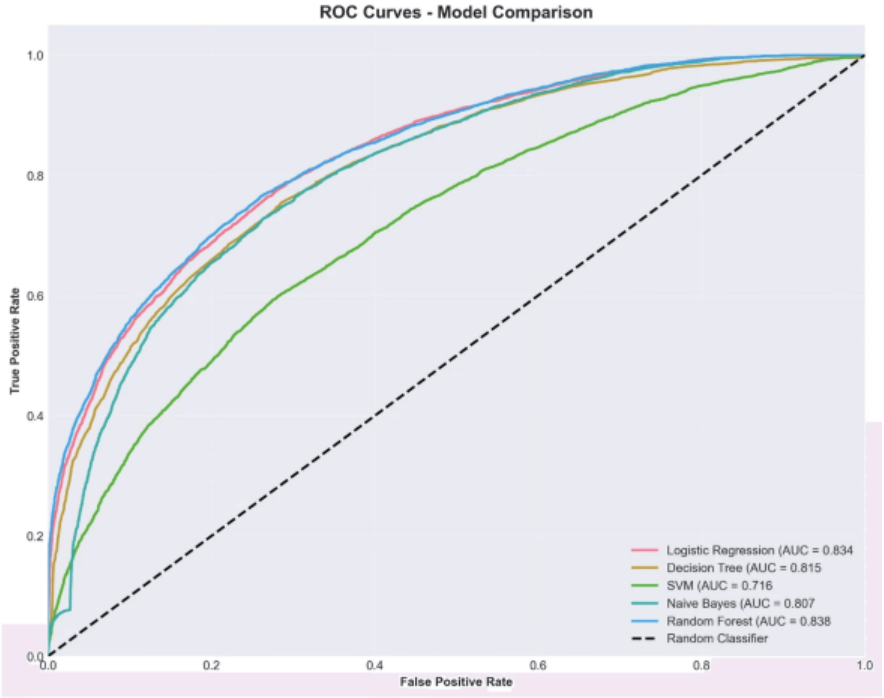
ROC Curve: Confirming Random Forest's Superiority

What is ROC Curve?

The ROC (Receiver Operating Characteristic) curve plots the True Positive Rate (Recall) against the False Positive Rate. A curve closer to the top-left corner indicates better model performance.

AUC Scores by Model

Model	AUC Score
Random Forest	0.838
Logistic Regression	0.834
Decision Tree	0.815
Naive Bayes	0.807
SVM	0.716



Statistical Justification

Random Forest achieves the highest AUC score (0.838), statistically confirming its superior ability to discriminate between employees who will leave and those who will stay, regardless of classification threshold.

Feature Importance: Identifying the Critical Retention Levers

*Attrition is primarily driven by factors related to **career progression, personal circumstances, and work environment**, not just income. Understanding these drivers enables targeted retention strategies.*

Rank	Feature	Importance Score	Business Implication
1	**Job Level**	**0.241**	Employees at lower levels or those with limited growth potential are most likely to leave. Clear career pathways and advancement opportunities are critical.
2	**Marital Status (Single)**	**0.165**	Single employees show significantly higher propensity to leave, suggesting greater mobility and fewer family ties anchoring them to the organization.
3	**Remote Work**	**0.126**	The presence or absence of remote work options is a major factor in retention. Flexible work arrangements are now a key employee expectation.
4	**Work-Life Balance**	**0.096**	Poor work-life balance is a direct and significant driver of turnover. Workload management and schedule flexibility are essential retention factors.
5	**Marital Status (Married)**	**0.072**	Married employees are less likely to leave than single employees, indicating that family responsibilities create stability and reduce job mobility.

Strategic Takeaway

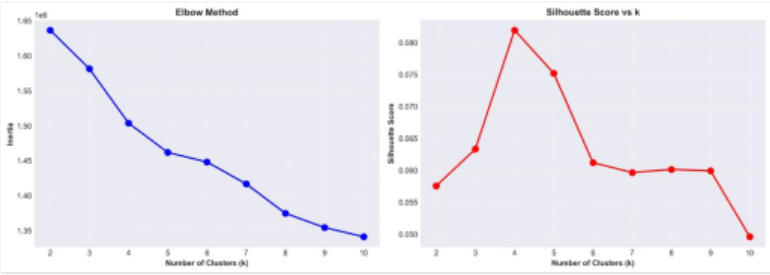
Retention strategies should prioritize career path clarity (Job Level) and flexible work arrangements (Remote Work, Work-Life Balance), as these are statistically the most influential factors driving employee attrition.

Personal circumstances (Marital Status) provide segmentation opportunities for customized HR interventions targeting different employee groups.

Unsupervised Learning: K-Means Clustering Methodology

Approach: K-Means Clustering was used to segment the employee population into distinct groups for targeted HR strategies. This unsupervised learning technique groups employees with similar characteristics together without requiring labeled attrition outcomes.

Determining Optimal Number of Clusters



Elbow Method and Silhouette Score analysis for k=2 to k=10

k = 3

Chosen Value

Clear elbow at k=3

Optimal for interpretability

Aligns with business needs

Optimal cluster selection

Cluster Determination Analysis

Elbow Method: Shows a clear 'bend' or elbow at **k=3**, suggesting that 3 clusters is a good number.

Silhouette Score: Reaches its peak at **k=4**, indicating the best-defined clusters (score ≈ 0.082).

Decision Rationale: We chose **k=3** for simplicity and clear interpretability, aligning with the strong visual evidence from the Elbow Method and business requirements for actionable segmentation.

K-Means Results: Workforce Segmentation and Application

PCA Visualization: Three Distinct Employee Clusters

✓The data clearly separates into three distinct, elongated clusters when visualized using Principal Component Analysis (PCA). The color gradient represents cluster membership, showing well-defined separation between groups. PC1 (6.3% variance) and PC2 (6.0% variance) effectively capture the primary dimensions of employee differentiation.

Employee Cluster Profiles

Cluster 1: High-Value/High-Risk

Characteristics

Senior-level employees with high expectations, strong performance, but elevated attrition risk due to external opportunities and career advancement demands.

Cluster 2: Stable/Mid-Career

Characteristics

Mid-level employees with moderate tenure, balanced work-life satisfaction, and lower attrition risk. Core workforce segment with retention stability.

Cluster 3: New/Low-Engagement

Characteristics

Entry-level or newer employees with limited tenure, lower engagement scores, and higher attrition probability. Early intervention critical.

Practical Application to HR Retention Strategy

These three clusters enable customized, segment-specific retention interventions. Each group has distinct characteristics, risk profiles, and HR needs, allowing for targeted resource allocation and strategy optimization.

→ **Cluster 1 (High-Value/High-Risk):** Executive mentorship, accelerated leadership development, competitive compensation reviews, and external networking opportunities to retain top talent.

→ **Cluster 2 (Stable/Mid-Career):** Career progression clarity, skills development programs, and work-life balance maintenance to sustain engagement and reduce unnecessary turnover.

Strategic Recommendations for Retention

Summary of Findings

- ✓ **Prediction:** Random Forest model (AUC=0.838, F1=0.765) provides reliable tool for identifying at-risk employees before they leave.
- ✓ **Key Drivers:** Top factors are Job Level, Marital Status (Single), and Remote Work options—not just salary.
- ✓ **Segmentation:** Workforce can be effectively segmented into 3 distinct groups for customized HR interventions.

1 **Career Pathing**

Implement clear, accelerated promotion tracks for employees in lower Job Levels. This addresses the #1 attrition driver and provides growth visibility.

2 **Flexibility & Work-Life Balance**

Review and expand remote work policies. Actively monitor and improve Work-Life Balance metrics through flexible schedules and workload management.

3 **Targeted Retention Program**

Use the predictive model to generate a weekly 'At-Risk' list. Assign HR business partners to proactively engage with flagged employees with personalized retention conversations.



References and Data Sources

Data Sources

Primary Dataset

Employee Attrition Dataset

74,610 records with 24 features including demographic, organizational, and psychological factors. Anonymized/Synthetic data for educational purposes in Big Data analytics.

Data Characteristics

Balanced classification problem (47.5% attrition, 52.5% retention). Comprehensive feature set enabling multi-variate analysis of attrition drivers.

Academic References

Machine Learning

Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5–32.

Data Mining

Han, J., Kamber, M., & Pei, J. (2012). *Data Mining: Concepts and Techniques* (3rd ed.). Elsevier.

Implementation

Scikit-learn. (2023). Machine Learning in Python. scikit-learn.org

Model Evaluation

Fawcett, T. (2006). ROC analysis. *Pattern Recognition Letters*, 27(8), 861–874.

Team Contributions

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Data Cleaning, EDA, Visualization Design

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Classification Models, Evaluation, K-Means Clustering

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Report Writing, Presentation, Statistical Analysis

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